# Running the ten-day mathematical diagnostic

Aleph Platinum · Version 1 · 17 October 2026

## What this can tell us

We want to know what mathematics the student can recall, explain, execute, recognize and combine. The same student may know a formula, miss a domain restriction, and solve a harder problem by reconstructing the formula. Preserve those differences.

This is a structured diagnostic prototype. It is not a normed psychometric instrument, an IQ test, an AMC/AIME score predictor or a complete survey of high-school mathematics. Difficulty labels and times are provisional. Geometry, trigonometry and probability are outside the current agreed scope. No pass/fail cutoff or overall ability score is justified.

All 133 questions are original. There are 25 four-level ladders, 25 parallel checks, five delayed confirmations, two additional last-digit variants and one optional Vieta-jumping probe. The ten daily common cores contain 80 questions. The remaining 53 are reserves; a student should attempt at most two selected reserves per day. They are not expected to solve all 133 questions.

## Before Day 1

Explain: “Some questions will be easy and others unfamiliar. Show us what you tried. An honest partial attempt is useful. We are choosing what to teach next, not assigning you an intelligence score.”

Use the same environment each day where feasible. Provide scratch paper and the current day PDF only. Do not distribute the combined student book or reserve book to the learner in advance. Keep the Day-10 changed questions unseen. The combined PDFs are administration/printing copies.

On the independent pass, use no calculator, notes, internet or AI. If an accommodation is needed, provide it and record the changed conditions. Do not treat a longer time allowance, typed versus handwritten work, interruption or language clarification as an equivalent timing condition.

Record prior experience: school curriculum; time since last mathematics study; familiarity with contest mathematics; whether the student has encountered rational-root arguments, Vieta formulas, functional equations or Vieta jumping. Do not convert unfamiliar terminology into a conceptual deficit without an ordinary-language follow-up.

## The 60-minute session

| Block | Maximum minutes | Procedure |
|---|---|---|
| Setup | 3 | Check conditions, explain the procedure, record interruptions or accommodations. |
| Common core | 30 | Eight questions in printed order. Work independently; move on if stuck. |
| Targeted follow-ups | 20 | Tutor selects at most two unused reserve questions, up to ten minutes each including any hint. |
| Log | 7 | Record confidence, familiarity, status and useful work; acknowledge effort without teaching the solutions. |

Stop at 60 minutes. If a block finishes early, allow checking work or move to the next block; do not add more items to fill time. Record actual duration. An unfinished core is valid diagnostic evidence, not homework to be completed later and folded into the independent score.

Item estimates are planning estimates, not time penalties. The common cores have estimated totals of 16–28 minutes. Actual times must be piloted. A student who does not reach an item has not demonstrated that they cannot solve it.

Some recognition questions appear before same-session recall/direct-use questions so that the test does not name the method immediately before asking the student to choose it. This changes administration order, not the meaning of the skill ladder. Day-1 recall and repeated testing may still prime later work; log familiarity rather than claiming a completely unprimed measurement.

## Selecting the two follow-ups

Read the written work. Identify one uncertainty worth resolving, not merely the lowest score. Record the core ID, the visible evidence and what the follow-up is intended to distinguish.

- **No usable setup / unfamiliar term:** choose that skill's -R parallel check. Rephrase terminology only after recording the independent attempt; record the clarification as a cue.
- **Sound setup, incorrect calculation:** choose -R to see whether the same reasoning works on changed numbers. Do not label an arithmetic slip a knowledge gap.
- **Accurate independent direct or recognition work:** choose that skill's -4 extension if it would tell us something useful.
- **Correct answer with invalid or absent reasoning:** ask the student to explain the work within the same item budget; a correct final number alone is insufficient evidence.
- **Unreached item:** do not infer a gap. If it concerns an important uncertainty, use a follow-up while recording that it was first attempted in the adaptive block.

Choose a second branch from a different uncertain skill where possible. Use only the reserve IDs listed in the day's tutor guide, within the 20-minute limit. A five-minute repair may be followed by a request to explain the same work, but not by extra unrecorded test questions.

Do not repeat an exact reserve already seen as independent corroboration. Select a different unused applicable question, or report insufficient evidence. Never improvise an unverified replacement and silently count it as a standardized item.

A failure to recall a result does not prohibit a reconstruction/application attempt. Conversely, knowing the result does not guarantee recognition. The ladder is a profile of claims, not an automatic rule that all lower rungs must be passed before attempting a higher one.

## Hints, feedback and learning

Do not offer content hints during the common core. An administrative clarification may identify the printed symbols or restate the task without naming a method; record it.

During a targeted follow-up, allow one printed neutral hint only after the independent attempt is recorded. Save both attempts and the hint wording/time. A correct result after a hint is **hinted-correct**, not independent-correct.

Prefer keeping worked answers unavailable until after Day 10. Between sessions, acknowledge that the work was recorded but avoid teaching the tested results. If the learner receives feedback, studies independently or sees a solution, record the skill, source and date. A later success can then show retention or learning, but cannot be described as stable knowledge present at baseline.

The optional EX-VJ question is a prior-exposure/enrichment probe. It combines Vieta's formulas, integer constraints and descent. Not having encountered Vieta jumping is not a baseline deficit. It should replace an adaptive slot, never add time or lower a foundational rating.

## Recording an attempt

Use the supplied attempt-record.csv or its equivalent. Keep the student's work linked to the item ID. Record:

- Outcome: independent-correct, partial, attempted-incorrect, not-attempted, not-reached, hinted-correct or hinted-incorrect.
- Reasoning: sound, incomplete, invalid or absent.
- Execution: accurate, slip or not-assessable.
- Confidence before feedback: 0 guess, 1 uncertain, 2 fairly sure, 3 sure.
- Familiarity: new, seen similar or seen exact.
- Hint/clarification text; approximate active time; answer/teaching exposure; scorer notes.

An item has three descriptive credit anchors in the solution guide. Award 0–3 using the visible work, accepting any equivalent valid method. A correct alternative proof does not need to match the author's trick. The anchors support item-level interpretation, not a total ability score. A method can be sound even if a copied number makes the final answer wrong; preserve that observation separately.

Do not invent a misconception from a blank response. “Not encountered,” “could not retrieve,” “could not choose a method,” and “made an invalid inference” require different evidence. The error lists are hypotheses to investigate, not automatic labels.

## Interpreting a skill ladder

For every skill, record evidence separately at L1 knowledge, L2 direct use, L3 recognition and L4 combination. Use these statuses:

- **Not checked:** no usable independent evidence.
- **Provisional success:** one independently sound response.
- **Corroborated success:** a second independently sound changed example probes the same claim; an exposed exact repeat does not qualify.
- **Uncertain / mixed:** partial or contradictory evidence requiring follow-up.
- **Observed difficulty:** visible work identifies a difficulty, with the remaining uncertainty stated. One failed item alone is not a stable trait.

Do not erase secure lower-level evidence after failure on an extension. Do not infer every lower skill from one hard success unless the written work actually demonstrates it. A learner who derives a forgotten formula has shown reconstruction even if unaided retrieval remains weak.

The common core gives one observation at each of the first three levels for each skill. Five delayed confirmations target selected claims; the reserves provide additional evidence where needed. This cannot corroborate every rung of every skill within ten hours. Leave uncorroborated claims marked provisional. More testing is useful only when it changes a teaching decision.

## Example interpretation — fictional, not learner data

Suppose a learner states Vieta's formulas correctly, reads sum/product correctly from a quadratic, but writes $r^2+s^2=(r+s)^2$. On the parallel check they need the printed hint before subtracting the cross term correctly.

Report: “Root sum/product knowledge and direct coefficient use were demonstrated once. The squared-sum identity was applied incorrectly. The parallel task was completed after a cue, so independent recognition remains uncertain. Begin with expanding $(r+s)^2$ and checking one numerical example; later use a fresh uncued expression.”

Do not report: “Algebra mastered,” “low ability,” or “knows Vieta at 70%.”

## The final learner report

Use skill-profile.csv. For each skill include the observed level evidence, item IDs, visible misconception or uncertainty, the next teaching action and a fresh check to confirm repair. The next action should name a concrete learning task, not “revise algebra.”

Group the report into: available knowledge; sound but slow methods; execution errors; method-selection difficulties; untested/uncorroborated areas; enrichment exposure. Include accommodations and any learning during the ten days. Keep confidence and familiarity separate from correctness.

## Pilot before stronger claims

Collect actual times, skip patterns, ambiguous-wording reports and scorer disagreements. Have another scorer independently score a sample of the same written work using the anchors, investigate disagreements and revise anchors. Have an independent mathematical reviewer inspect the bank. Reassess difficulty labels using student responses, check whether items discriminate intended skills, and check whether prior contest exposure or language demands dominate performance.

Do not fit an IRT/Rasch model, publish percentiles, impose a calibrated pass mark or claim reliability from this single learner's ten-day run. Those require a suitable dataset and a justified measurement model.

## Review status

The content was drafted, corrected and released day by day using question-only calculation checks, comparison with authored solutions, author proof/language review and PDF layout checks. General proofs received same-author review. Finite sample checks of identities are sanity checks, not proofs. The release is labelled AUTHOR_REVIEWED_COMPUTATION_CHECKED. Independent editorial review and psychometric calibration have not been performed.

Design references: ETS evidence-centered design, MAA domains, MAA invitational competitions. These references inform the design, not validate this particular test.